

## Graphical Abstract

### **Hand Gesture Recognition Using Convolutional Neural Networks for Reliable Human–Computer Interaction**

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## Highlights

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- Developed a lightweight CNN for automatic recognition of hand gestures.
- Utilized approximately 13,000 RGB gesture images from the Kaggle dataset.
- Achieved validation accuracy of approximately 99.15
- Confusion matrix, precision, recall and F1-score are used to evaluate the performance
- Demonstrated reliable human-computer interaction via computer vision. Convolutional Neural Networks for Robust Hand Gesture Recognition in Human-Computer Interaction

# Hand Gesture Recognition Using Convolutional Neural Networks for Reliable Human–Computer Interaction

Mohammed Aiman Sathik<sup>a</sup>

<sup>a</sup>*Department of Business, University of Europe for Applied Sciences, Think Campus,  
Konrad-Zuse-Ring 11, Potsdam, 14469, Brandenburg, Germany*

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## Abstract

Hand gesture recognition is one of the fast growing research areas in computer vision, which facilitates natural communication between humans and intelligent computer systems without physical contact with traditional input devices. Gesture-based interfaces are increasingly used in robotics, healthcare, gaming, smart homes, industrial automation, virtual reality and autonomous systems where traditional interaction methods are inconvenient or inefficient. Deep learning has significantly improved the accuracy of image classification, but many gesture recognition systems currently use computationally expensive transfer learning models that demand considerable processing power and memory. So, designing a compact custom Convolutional Neural Network which can achieve high classification accuracy with low computational cost is still a challenging research problem. In this paper, a lightweight Convolutional Neural Network framework implemented in TensorFlow and Keras for automated hand gesture recognition is proposed. The training and validation datasets were obtained by resizing 13,000 RGB images labelled in the publicly available Kaggle Hand Gesture Recognition Dataset to  $128 \times 128$  pixels, normalization and split into training and validation datasets in an 80:20 ratio. The proposed CNN architecture consists of three convolutional blocks followed by max-pooling, dropout regularization, fully connected layers and a Softmax output layer. Experimental performance was evaluated by validation accuracy, precision, recall, F1-score, validation loss, confusion matrix analysis, classification report and qualitative prediction visualisation. The developed CNN got validation accuracy of about 99.15, while the precision, recall, F1-score got about 0.99 for all gesture classes. The confusion matrix showed very few classification errors, indicating excellent model generalization. The proposed framework shows

that a well-designed lightweight CNN can reach excellent gesture recognition performance without resorting to computationally intensive transfer-learning architectures. The developed system offers a practical foundation for future real-time gesture recognition applications, enabling reliable and intuitive human-computer interaction. Hand Gesture Recognition, Convolutional Neural Network, Computer Vision, Deep Learning, TensorFlow, Human-Computer Interaction

*Keywords:*

Hand Gesture Recognition, Convolutional Neural Network, Computer Vision, Deep Learning, TensorFlow, Human-Computer Interaction

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## 1. Introduction

Hand gesture recognition has become one of the fastest growing research areas in computer vision and artificial intelligence as it provides natural communication between humans and intelligent systems without having to rely on traditional input devices such as keyboards, computer mice or touch screens. Recent advances in deep learning, especially Convolutional Neural Networks (CNNs), have greatly improved the capacity of machines to automatically recognize complex visual patterns from digital images. Consequently, gesture recognition systems have become more and more important in robotics, healthcare, industrial automation, virtual reality, gaming, smart homes, and intelligent human-computer interaction.

Deep learning has revolutionized image classification by allowing neural networks to automatically learn hierarchical visual features directly from image data. Compared to the traditional machine learning methods which require hand-crafted feature extraction, CNNs provide higher recognition accuracy, better robustness and better generalization over different imaging conditions. These advantages have made CNN-based image classification one of the most widely adopted approaches for visual recognition tasks. However, designing an efficient and computationally lightweight gesture recognition system remains an important research challenge despite these advances.

Many existing studies employ large transfer-learning architectures that achieve excellent classification performance but require a lot of computational resources, so they are not suitable for real-time deployment on reconstrained nodes. Thus, it has become an active research area to develop light-weight CNN models that can maintain a high predictive performance while mini-

mizing computational complexity. This study proposes a lightweight Convolutional Neural Network framework for automatic hand gesture recognition using publicly available Kaggle Hand Gesture Recognition Dataset.

The suggested methodology involves image pre-processing, CNN model development, supervised learning, and detailed performance evaluation based on validation accuracy, precision, recall, F1-score, confusion matrix analysis, and classification reports. The remaining part of this section explains the application background, machine learning context, research gap, objectives, contributions, and overall organization of the study.

### *1.1. Problem Statement*

Automatic hand gesture recognition is still a challenging computer vision problem as the appearance of the gestures can vary significantly due to the variations in illumination, hand orientation, background complexity, camera viewpoint and the individual user characteristics. These variations often decrease classification accuracy and restrict the practical deployment of gesture recognition systems in real-world environments.

Although deep learning has significantly improved image classification performance, many existing gesture recognition systems depend on large-transfer learning models that need extensive computational resources. These models might not be suitable for applications that require fast inference and deployment on resource-limited hardware.

Therefore, there is a need to develop a lightweight CNN architecture which can accurately recognize hand gestures while keeping the computational efficiency. The problem addressed in this study is the design and evaluation of a custom Convolutional Neural Network for recognizing five different hand gesture classes using RGB images obtained from the Kaggle Hand Gesture Recognition Dataset. The aim of the proposed system is to achieve high classification accuracy with reduced computational complexity and suitability for future real time human computer interaction applications.

### *1.2. Application Background and Practical Importance*

Hand gesture recognition has become an increasingly important research area in computer vision, as it allows for intuitive and touch-free communication between humans and intelligent computer systems. Instead of with generic input devices such as keyboard, computer mice, or touch screens, users can naturally interact with hand gestures captured by standard RGB cameras. Interaction techniques such as these are especially useful in situations where

62 physical contact is inconvenient, dangerous or not feasible. The applica-  
 63 tion domain of this work covers a number of industries such as healthcare,  
 64 robotics, industrial automation, virtual reality, augmented reality, gaming,  
 65 smart homes, automotive systems and assistive technologies. In healthcare,  
 66 surgeons and medical staff can control imaging systems from a distance, with-  
 67 out having to touch contaminated equipment, thus reducing the risk of in-  
 68 fection. Also, in rehabilitation systems, which monitor patient exercises and  
 69 improve physical therapy, gesture recognition is widely used. In industrial  
 70 environments, workers wearing protective gear can control machinery using  
 71 gesture commands, eliminating the need for physical control panels. Like-  
 72 wise, gesture-based interfaces increase the accessibility for elderly users and  
 73 persons with physical disabilities by offering a more natural way of communi-  
 74 cation with intelligent systems. Conventional human-computer interaction is  
 75 largely dependent on manual input devices such as keyboards, touchscreens,  
 76 joysticks and remote controllers. Although such interfaces are efficient in  
 77 many applications, they need physical contact and are often not suitable  
 78 in environments where hygiene, mobility or user accessibility are important  
 79 concerns. Moreover, manual interaction can decrease operational efficiency  
 80 in situations where quick decision making or hands-free control is required.  
 81 Wearable sensor-based gesture recognition systems have also been studied;  
 82 however, such methods require additional hardware, raise the cost of im-  
 83 plementation, and often decrease the user comfort. An appealing option is  
 84 automated image analysis by deep learning, since up-to-date Convolutional  
 85 Neural Networks (CNNs) are able to extract meaningful visual features au-  
 86 tonomously from RGB images. In contrast to traditional image-processing  
 87 techniques that use manual feature extraction, CNNs show better resilience  
 88 to changes in illumination, background complexity, hand orientation, and  
 89 camera viewpoint. The availability of cheap cameras and the advances in  
 90 deep learning has made vision based gesture recognition feasible and eco-  
 91 nomically viable for real world applications. The rapid proliferation of intel-  
 92 ligent devices makes gesture recognition technologies even more important.  
 93 Industry reports report that Today, there are 4 billion smart devices with  
 94 embedded cameras deployed worldwide, and as a result, there is a huge need  
 95 for reliable touch-free interaction systems. Thus, the lightweight and com-  
 96 putationally efficient CNN100 models with high recognition accuracy can be  
 97 an important research direction for future intelligent human-computer inter-  
 98 action systems. Fig. 1 shows a few representative application areas where  
 99 automatic hand gesture recognition can improve user interaction and opera-

100 tional efficiency. Figure 1: Example application areas of vision-based hand  
 101 gesture recognition such as healthcare, robotics, smart homes, gaming, and  
 102 industrial automation. Investigate and ready the chosen Kaggle hand gesture  
 103 recognition dataset To design a light weight Convolutional Neural Network  
 104 (CNN) for classification of five hand gesture categories. For training the pro-  
 105 posed CNN with supervised deep learning techniques To evaluate the model  
 106 using validation accuracy, precision, recall, F1 score, confusion matrix and  
 107 classification report. The training behavior and overall performance of the  
 108 proposed CNNs are analyzed. To evaluate the applicability of the developed  
 109 model for future real-time human-computer interaction applications.

110 Figure 1 illustrates several representative application areas where auto-  
 111 matic hand gesture recognition can improve user interaction and operational  
 112 efficiency.

- 113 • To analyze and preprocess the selected Kaggle Hand Gesture Recogni-  
 114 tion Dataset.
- 115 • To develop a lightweight Convolutional Neural Network (CNN) for clas-  
 116 sifying five hand gesture categories.
- 117 • To train the proposed CNN using supervised deep learning techniques.
- 118 • To evaluate the model using validation accuracy, precision, recall, F1-  
 119 score, confusion matrix, and classification report.
- 120 • To analyze the training behaviour and overall performance of the pro-  
 121 posed CNN.
- 122 • To assess the suitability of the developed model for future real-time  
 123 human-computer interaction applications.

### 124 1.3. Machine Learning and Computer Vision Context

125 Machine learning has revolutionized computer vision, making it possi-  
 126 ble to learn meaningful patterns automatically from digital images, without  
 127 the need for manually-designed feature extraction methods. In image clas-  
 128 sification tasks, machine learning algorithms analyse pixel-level information  
 129 and identify visual characteristics that discriminate an object category from  
 130 another. The features for hand gesture recognition are the hand shape, fin-  
 131 ger location, palm orientation, texture, ledges, and the spatial relationship

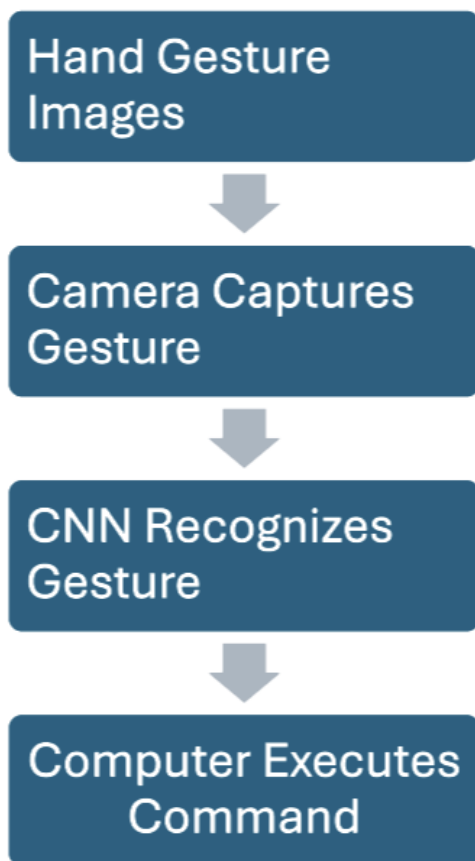


Figure 1: Representative application areas of vision-based hand gesture recognition including healthcare, robotics, smart homes, gaming, and industrial automation.

132 between different parts of the hand. Among the different deep learning tech-  
133 niques, Convolutional Neural Networks (CNNs) have been the most widely  
134 used architecture for image analysis because of their ability to automatically  
135 extract hierarchical visual features directly from raw images. Different from  
136 classical machine learning methods based on handcrafted descriptors like  
137 Histogram of Oriented Gradients (HOG), Scale-Invariant Feature Transform  
138 (SIFT) or Local Binary Patterns (LBP), CNNs learn low-level and high-level  
139 image representations in the learning process. Features such as edges and  
140 corners in Simple134 are detected in early convolutional layers, while deeper  
141 layers are responsible for recognising more complex structures such as finger  
142 arrangements and complete hand gestures. CNNs have several important



143 advantages for the proposed hand gesture recognition system. First, auto-  
 144 matic feature extraction removes the need for image engineering, reducing  
 145 development complexity and improving model generalisation. Second, CNNs  
 146 can identify gestures patterns under moderate changes of illumination and  
 147 background camera angle and hand position. Third, their hierarchical learn-  
 148 ing capability enables accurate classification of the five gesture categories  
 149 used in this study: Backward, Forward, Left, Right, and Unknown.<sup>144</sup> Deep  
 150 learning is especially suitable for this project because the Kaggle Hand Ges-  
 151 ture Recognition Dataset contains thousands of labeled RGB images that  
 152 allow supervised learning of gesture representations. During training the  
 153 CNN learns progressively discriminative visual patterns associated with each  
 154 gesture class, and uses this knowledge to classify previously unseen images.  
 155 This ability to generalise beyond the training dataset makes CNNs highly ef-  
 156 fective for practical gesture recognition applications. However, deep learning  
 157 models require sufficient labelled data to obtain high predictive performance.  
 158 The limited datasets may increase the risk of overfitting, where the model  
 159 memorises training samples instead of learning general visual characteristics.  
 160 To address this problem, the proposed framework uses image normalisation,  
 161 dataset shuffling and an 80:20 training-validation split . This allows the  
 162 CNN to learn robust gesture features, while keeping good generalisation per-  
 163 formance on unseen data .

#### 164 *1.4. Transfer Learning for the Selected Problem*

165 One of the most widely adopted approaches in deep learning is transfer  
 166 learning, which allows neural networks to reuse knowledge learned from very  
 167 large image datasets such as ImageNet. On the other hand, transfer learning  
 168 does not train all network parameters from random initialization, but starts  
 169 with a model that has already learned general visual features from millions  
 170 of images. These pre-trained models can then be fine-tuned to a new im-  
 171 age classification task with much less training samples and shorter training  
 172 times. Training a deep Convolutional Neural Network from scratch requires  
 173 a large labelled dataset and great computational resources. When there are  
 174 only a few training images, the randomly initialized models are prone to  
 175 overfitting, unstable convergence, and lower classification accuracy. Transfer  
 176 learning offers a solution to these challenges by providing a good starting set  
 177 of feature representations that improve learning efficiency and model gener-  
 178 alisation. Pre-trained networks learn universal visual features during their  
 179 initial training. The early layers respond to edges, textures, colours and ba-

180 sic geometric patterns. The deeper layers respond to more complex object  
 181 structures. These learned representations are reused for the target applica-  
 182 tion via feature extraction or fine-tuning during transfer learning. Feature  
 183 extraction consists of training only the last classification layers while keeping  
 184 most of the pre-trained layers fixed, while fine-tuning allows the update of  
 185 selected layers using the new dataset. Popular transfer-learning architec-  
 186 tures include VGG16, ResNet50, MobileNetV2, DenseNet, and EfficientNet.  
 187 These models have shown great performance on a wide range of computer  
 188 vision applications as they combine powerful feature extraction and strong  
 189 classification capability. However, they generally require more computational  
 190 resources than light-weight custom CNN architectures. The main focus of  
 191 this study is the designing of a custom Convolutional Neural Network for five-  
 192 class hand gesture recognition, but transfer learning provides an important  
 193 benchmark for comparison. Lightweight CNN models are still attractive since  
 194 they need less parameters, shorter training time, less memory consumption  
 195 with high recognition accuracy. This work explores if a custom CNN, prop-  
 196 erly designed, can achieve accurate gesture recognition performance without  
 197 the need for computationally expensive transfer-learning architectures.

### 198 *1.5. Explainable AI and Trustworthiness*

199 With increasing complexity of deep learning models, understanding the  
 200 basis of their predictions has become an important aspect of trustworthy  
 201 Artificial Intelligence (AI) [199, 200]. Accuracy is a typical measure for eval-  
 202 uating model performance. However, accuracy alone does not ensure that a  
 203 model has learned meaningful visual features. A CNN might be very ac-  
 204 curate by relying on irrelevant background information, lighting conditions  
 205 or other dataset-specific characteristics, instead of the actual hand gesture.  
 206 Such behaviour is known as background bias or short-cut learning and can  
 207 significantly reduce model reliability when applying to real-world environ-  
 208 ments. For hand gesture recognition, it is important to ensure that the CNN  
 209 focuses on the hand region, finger positions, palm orientation, and gesture  
 210 shape instead of the surrounding objects or backgrounds. If the model learns  
 211 the wrong visual cues, its predictions can become unreliable when images are  
 212 taken in different lighting conditions, camera angles or backgrounds. There-  
 213 fore, evaluating only numerical performance metrics, such as accuracy, 8  
 214 precision, recall and F1-score are not sufficient for evaluating the trustwor-  
 215 thiness of a deep learning model. Explainable Artificial Intelligence (XAI)  
 216 offers techniques that enhance the transparency of deep learning models by

217 revealing the image regions that most strongly contribute to a prediction.  
 218 One popular method is Gradient-weighted Class Activation Mapping (Grad-  
 219 CAM), which generates220 heatmaps showing the areas of an image that con-  
 220 tribute to the CNN’s decision. These visual explanations allow researchers  
 221 to verify if the network is focusing on pertinent hand regions instead of irrel-  
 222 evant223 background features. A further important explainability method is  
 223 SHapley Additive exPlanations (SHAP). SHAP assigns positive or negative  
 224 importance values to estimate the contribution of individual image regions to  
 225 the final prediction. Unlike Grad-CAM, which mainly provides spatial atten-  
 226 tion maps, SHAP allows for a detailed interpretation of feature importance  
 227 and a comparison between correctly and incorrectly classified samples. Al-  
 228 though Grad-CAM and SHAP were not implemented in the present project,  
 229 they are valuable techniques for future model validation and improvement.  
 230 The integration of explainable AI in hand gesture recognition systems can  
 231 help improve users’ trust and model transparency and help developers iden-  
 232 tify potential biases or weaknesses in the trained CNN. Such interpretabil-  
 233 ity is important for applications involving healthcare, industrial automation,  
 234 assistive technologies and other safety-critical human-computer interaction  
 235 systems where reliable and understandable predictions are paramount.

### 236 *1.6. Research Gap*

237 The effectiveness of deep learning for hand gesture recognition has been  
 238 proved by many studies, but there are still some research challenges.243 Most  
 239 of the existing systems are based on large transfer learning architectures like  
 240 VGG16, ResNet50, EfficientNet and DenseNet that achieve excellent classi-  
 241 fication accuracy but require hugecomputational resources and long training  
 242 times. These models are often not deployable on mobiledevices, embedded  
 243 platforms or other resource-constrained systems. Moreover, many published  
 244 studies focus mainly on maximizing clasification accuracy with little dis-  
 245 cussion on computational249 efficiency, model complexity and deployment  
 246 feasibility. Relatively few studies investigate if lightweight custom CNN ar-  
 247 chitectures 9 can achieve similar performance without pre-trained networks.  
 248 Another limitation is that many studies report only overall accuracy, with  
 249 lack of sufficient analysis using precision, recall, F1-score, confusion matri-  
 250 ces,254 and class-wise performance metrics. The present study addresses  
 251 these limitations by proposing a lightweight Convolutional Neural Network  
 252 for five-class hand gesture recognition. The comprehensive experimental eval-  
 253 uation keeps the high predictive performance of the proposed framework in

254 the while emphasizing computational simplicity.

255 The present study addresses these limitations by proposing a lightweight  
256 Convolutional Neural Network for five-class hand gesture recognition. The  
257 proposed framework emphasizes computational simplicity while maintaining  
258 high predictive performance through comprehensive experimental evaluation.

### 259 1.7. Research Questions

260 The following research questions guide the development and evaluation  
261 of the proposed hand gesture recognition framework.

- 262 • **RQ1:** Can a lightweight Convolutional Neural Network accurately clas-  
263 sify five different hand gesture categories using RGB images?
- 264 • **RQ2:** How effective are image preprocessing techniques such as resizing  
265 and normalization in improving CNN performance?
- 266 • **RQ3:** Does the proposed CNN achieve reliable classification perfor-  
267 mance across all gesture classes?
- 268 • **RQ4:** Can the proposed lightweight CNN provide competitive perfor-  
269 mance without relying on computationally expensive transfer learning  
270 architectures?
- 271 • **RQ5:** Which evaluation metrics best demonstrate the effectiveness of  
272 the proposed hand gesture recognition system?

### 273 1.8. Contributions and Novelty

274 The major contributions of this research are summarized as follows.

- 275 • Development of a lightweight Convolutional Neural Network for auto-  
276 matic five-class hand gesture recognition.
- 277 • Implementation of a complete image preprocessing pipeline including  
278 resizing, normalization, dataset partitioning, and supervised learning  
279 using TensorFlow and Keras.
- 280 • Comprehensive experimental evaluation using validation accuracy, pre-  
281 cision, recall, F1-score, confusion matrix analysis, classification report,  
282 and qualitative prediction results.

- Demonstration that a custom CNN architecture can achieve excellent classification performance while maintaining lower computational complexity than many transfer learning approaches.
- Provision of a practical framework that may support future touch-free human–computer interaction systems in healthcare, robotics, industrial automation, smart environments, and assistive technologies.

### 1.9. Report Organization

The rest of this paper is organized as follows. Section 2 reviews the related literature to hand gesture recognition, deep learning, Convolutional Neural Networks, transfer learning, image preprocessing, and Explainable Artificial Intelligence. The proposed methodology, dataset, preprocessing techniques, CNN architecture, training procedure, and evaluation metrics are described in Section 3. Section 4 presents the experimental results obtained from the proposed model as well as quantitative and qualitative performance analysis. Section 5 discusses the results, practical implications, limitations, and future research directions. Finally, Section 6 concludes the study, summarizing the major contributions and highlighting opportunities for future work.

## 2. Literature Review

### 2.1. Image-Based Machine Learning in the Application Domain

Hand gesture recognition has become an active research area in computer vision since it enables intuitive communication between humans and intelligent systems through visual analysis of hand movements. Vision-based gesture recognition systems, such as Image-306, use digital cameras to capture RGB images and employ machine learning algorithms to recognize pre-defined gesture classes. Compared to wearable sensor-based approaches, vision-based methods are more cost-effective, easier to deploy, and do not require any additional hardware, making them suitable for applications in healthcare, robotics, smart homes, industrial automation, virtual reality, and assistive technologies. Early hand gesture recognition systems employed traditional image processing techniques, including skin colour segmentation, contour extraction, edge detection, and handcrafted feature descriptors. These methods performed well in controlled scenarios but often failed to cope with illumination variations, background complexities, camera viewpoint and hand orientation variations. Their performance also depended heavily on

manually engineered features, which restricted their generalization to new datasets. Machine learning algorithms such as Support Vector Machines (SVM), Decision Trees, Random Forests, and Artificial Neural Networks enhanced gesture recognition performance by learning discriminative patterns from labelled data. However, these approaches still depended on hand-crafted image descriptors, such as Histogram of Oriented Gradients (HOG), Local Binary Patterns (LBP), and Scale-Invariant Feature Transform (SIFT). Designing effective hand-crafted features required significant domain expertise, and often resulted in a decline in model robustness in real-world scenarios. Recent breakthroughs in deep learning have transformed image-based gesture recognition by enabling Convolutional Neural Networks (CNNs) to learn hierarchical visual representations automatically from raw image data. Instead of relying on manually extracted features, CNNs automatically learn to recognize edges, textures, shapes, and increasingly complex gesture features during training. This automatic feature learning ability has greatly improved recognition accuracy while reducing dependence on manual feature engineering. CNN-based image classification has become the dominant approach for modern hand gesture recognition systems as a result of its high predictive performance, scalability, and suitability for practical human–computer interaction applications.

## 2.2. CNN-Based Methods for Hand Gesture Recognition

Convolutional Neural Networks (CNNs) are the most widely adopted deep learning architectures in image classification as they learn visual features hierarchically from raw image data. Unlike traditional machine learning methods which depend on hand-designed image descriptors, CNNs combine feature extraction and classification into a single end-to-end learning framework. This ability has greatly improved recognition accuracy for a wide variety of computer vision applications, such as object detection, facial recognition, medical image analysis, and hand gesture recognition. A typical CNN consists of multiple convolution layers, pooling layers, activation functions and fully connected layers. Convolutional layers learn local image features like edges, corners and textures, and pooling layers down-sample features and enhance computational efficiency. As the network becomes deeper, it automatically learns higher-level features that represent the finger positions, hand orientation, and gesture shape. Finally, these learned features are combined by fully connected layers to classify the input image into predefined gesture categories. Several researchers have successfully used CNNs for hand gesture

354 recognition. Early CNN models showed that deep learning could significantly  
 355 outperform traditional classifiers using hand crafted features. More recent  
 356 works have proposed deeper architectures that can achieve excellent classifi-  
 357 cation accuracy even under varying illumination, background complexity, and  
 358 camera viewpoints. These studies confirm that CNNs provide more robust-  
 359 ness and better generalization compared to traditional machine learning tech-  
 360 niques. CNN-based gesture recognition has been successfully used in many  
 361 real-world applications such as robotic control, virtual reality, augmented  
 362 reality, healthcare monitoring, industrial automation, and assistive technole-  
 363 gies. CNNs can learn discriminative visual representations automatically to  
 364 recognize complex gesture patterns while minimizing the need for manual  
 365 feature engineering. However, despite the strong performance, deep CNN  
 366 models often require considerable computational resources and large train-  
 367 ing datasets. Thus, lightweight CNN architectures have received increasing  
 368 attention due to their ability to achieve competitive classification accuracy  
 369 while reducing computational complexity and memory requirements. The  
 370 present study adopts this approach to develop a lightweight custom CNN  
 371 model specifically for five-class hand gesture recognition using RGB images  
 372 from the Kaggle Hand Gesture Recognition Dataset.

### 373 *2.3. Transfer Learning Architectures*

374 Transfer learning has become one of the most popular approaches in com-  
 375 puter vision as it enables the reuse of knowledge learned from large-scale  
 376 image datasets such as ImageNet by deep learning models. Instead of train-  
 377 ing a deep neural network from scratch, researchers fine-tune or use feature  
 378 extraction with pre-trained models for new image-classification problems.  
 379 This approach reduces the training time considerably 13 improving classifi-  
 380 cation performance, especially when only limited labelled datasets are avail-  
 381 able. Several transfer-learning architectures have been successfully applied  
 382 to hand gesture recognition. VGG16 is popular due to its simple structure  
 383 and its strong ability of feature extraction, while ResNet50 contains residual  
 384 connections that enable training much deeper networks in an efficient way.  
 385 MobileNetV2 aims to be computationally efficient by using depthwise separa-  
 386 ble convolutions, making it suitable for mobile and embedded applications.  
 387 DenseNet121 facilitates feature reuse via dense connections, and Efficient-  
 388 Net optimizes network depth, width, and input resolution to achieve high  
 389 predictive accuracy with relatively low computational cost. These architec-  
 390 tures have been evaluated in previous studies on datasets with varying sizes

391 and class structures. Public datasets of hand gestures usually include from  
 392 a few thousand to tens of thousands RGB images of static or dynamic ges-  
 393 tures. Most of the datasets have between five to twenty gesture categories but  
 394 larger datasets with more gesture classes have also been reported. The pre-  
 395 dictive performance of deep learning models is heavily influenced by dataset  
 396 diversity, image quality, background variation, and class balance. Most pub-  
 397 lished studies perform image preprocessing prior to model training. Typical  
 398 preprocessing strategies involve resizing images, normalizing pixels, shuffling  
 399 datasets, augmenting data, and splitting data into training and validation  
 400 sets. Image augmentation techniques such as rotation, zoom, translation,  
 401 brightness adjustment, and flipping increase model robustness by increasing  
 402 the diversity of the dataset and reducing overfitting. Reported classifica-  
 403 tion performance for transfer-learning models is generally high, with many  
 404 studies achieving validation accuracies above 95 percent under controlled ex-  
 405 perimental conditions. However, performance varies significantly depending  
 406 on dataset complexity, image quality, preprocessing strategy and network  
 407 architecture. 3. Transfer-learning models have several limitations, despite  
 408 their excellent predictive performance. These models typically require signif-  
 409 icantly more computational resources, longer training time, and greater mem-  
 410 ory consumption than lightweight custom CNN architectures. Furthermore,  
 411 many published studies focus mainly on the overall classification accuracy,  
 412 with limited discussion on computational efficiency, deployment feasibility,  
 413 explainability and real-world robustness. These methodological issues that  
 414 remain unresolved motivate us to develop lightweight CNN models that  
 415 keep high recognition accuracy and reduce computational complexity. The  
 416 current work investigates the possibility of a customized CNN architecture for  
 417 reliable five-class hand gesture recognition on a simple preprocessing pipeline  
 418 and detailed performance evaluation.

#### 419 *2.4. Data Preprocessing and Image Augmentation*

420 A basic step of data preprocessing is required for deep learning, as the  
 421 quality of input images has direct impact on the learning ability of Con-  
 422 volutional Neural Networks. Proper preprocessing makes the dataset more  
 423 standardized, reduces unnecessary variations and enables the model to learn  
 424 meaningful visual features more effectively. Preprocessing also speeds up  
 425 the convergence in the training and improves the generalization capability of  
 426 the developed model in the image classification tasks. Several preprocessing  
 427 techniques have been widely used in previous hand gesture recognition stud-



ies. Image resizing is one of the most common procedures, since the input images of CNNs must be of the same dimension. Resizing guarantees that all samples have the same spatial resolution, while also reducing the computational needs during training. Another important preprocessing technique is pixel normalization, where the pixel values are scaled to the range of 0 to 1. This normalization improves the numerical stability during the optimization, and speeds up the convergence of the model. Image augmentation can also be used by many researchers to artificially increase the diversity of the dataset. The common augmentation methods are image rotation, horizontal and vertical translation, zooming, brightness adjustment, random cropping and horizontal flipping. These transformations enhance the visual variation of CNN, which aids in reducing overfitting and improving the robustness of the model to changes in illumination, viewing angle and hand orientation. Nevertheless, augmentation strategies should be selected with care as too many transformations may change the semantic meaning of a gesture and introduce wrong labels. Another important preprocessing step is the dataset partitioning. Most published studies have divided the data into training, validation and testing subsets to assess model performance objectively. Training data are used for model learning, validation data to track learning behaviour and hyper-parameter selection, while testing data provide an unbiased measure of predictive performance on previously unseen images. In this work, the Kaggle Hand Gesture Recognition Dataset was resized to 128 x 128 pixels, normalized, shuffled, and split into training and validation sets with a ratio of 80:20. No additional augmentation techniques were employed, as in many previous studies because the existing dataset already had sufficient variation within the five gesture classes. The preprocessing pipeline provided standardized input data for the proposed lightweight CNN while keeping computational efficiency and minimizing unnecessary processing overhead. In general, previous research consistently demonstrates that appropriate preprocessing strategies improve CNN performance by reducing noise, improving feature consistency, and enhancing model generalization. Thus, preprocessing remains one of the most important stages in building reliable image-based hand gesture recognition systems.

## 2.5. Model Evaluation and Performance Metrics

Model evaluation is an important part of deep learning as it determines the quality of a trained Convolutional Neural Network on never-seen-before data. Overall classification accuracy provides an important first indication

465 of model performance but can lead to misleading conclusions if used in isolation,  
 466 especially when datasets are characterized by class imbalance or varying  
 467 levels of difficulty of classification. Therefore, various evaluation metrics  
 468 are used in modern image-classification studies to comprehensively evaluate  
 469 the predictive performance. Classification accuracy is the ratio of correctly  
 470 classified samples to all evaluated images. High accuracy generally indicates  
 471 good learning, but does not provide information about the model’s  
 472 behaviour for individual classes. Accuracy is usually reported with precision,  
 473 recall and F1-score for a more balanced evaluation. The precision is  
 474 used to measure the proportion of correctly classified samples among the  
 475 predicted samples, indicating the reliability of a positive prediction. Recall  
 476 measures the proportion of actual samples that are correctly identified by  
 477 the model, reflecting its ability to detect all relevant gesture classes. The  
 478 F1-score is a single metric combining precision and recall by computing their  
 479 harmonic mean, providing a balanced measure of classification performance.  
 480 The confusion matrix is another popular evaluation tool because it shows  
 481 class-wise prediction results in a tabular form. Correct predictions are on  
 482 the main diagonal, and misclassified samples are represented by off diagonal  
 483 elements. This visualisation allows researchers to find common confusions  
 484 among gesture classes and to better analyse classification errors. Several  
 485 recent studies also report Receiver Operating Characteristic (ROC) curve  
 486 and the Area Under the Curve (AUC) for evaluation of classifier’s discrimination  
 487 ability. The ROC-AUC is particularly useful for binary classification  
 488 problems, but extensions to multi-class problems have also been used in deep  
 489 learning research. Additional studies include Precision–Recall curves Cohen’s  
 490 Kappa coefficient, and Matthews Correlation Coefficient to further evaluate  
 491 model robustness. The proposed lightweight CNN was evaluated using validation  
 492 accuracy, validation loss, precision, recall, f1-score, confusion matrix  
 493 analysis, classification report, and qualitative prediction visualization. The  
 494 use of multiple evaluation metrics provides a more comprehensive assessment  
 495 of model reliability than the overall accuracy alone and allows a thorough  
 496 analysis of the strengths and limitations of the developed gesture recognition  
 497 framework.

## 498 *2.6. Explainable Artificial Intelligence in Image Classification*

499 Model evaluation is an important part of deep learning as it determines  
 500 the quality of a trained Convolutional Neural Network on never-seen-before  
 501 data. Overall classification accuracy provides an important first indication

of model performance but can lead to misleading conclusions if used in isolation, especially when datasets are characterized by class imbalance or varying levels of difficulty of classification. Therefore, various evaluation metrics are used in modern image-classification studies to comprehensively evaluate the predictive performance. Classification accuracy is the ratio of correctly classified samples to all evaluated images. High accuracy generally indicates good learning, but does not provide information about the model’s behaviour for individual classes. Accuracy is usually reported with precision, recall and F1-score for a more balanced evaluation. The precision is used to measure the proportion of correctly classified samples among the predicted samples, indicating the reliability of a positive prediction. Recall measures the proportion of actual samples that are correctly identified by the model, reflecting its ability to detect all relevant gesture classes. The F1-score is a single metric combining precision and recall by computing their harmonic mean, providing a balanced measure of classification performance. The confusion matrix is another popular evaluation tool because it shows class-wise prediction results in a tabular form. Correct predictions are on the main diagonal, and misclassified samples are represented by off diagonal elements. This visualisation allows researchers to find common confusions among gesture classes and to better analyse classification errors. Several recent studies also report Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC) for evaluation of classifier’s discrimination ability. The ROC-AUC is particularly useful for binary classification problems, but extensions to multi-class problems have also been used in deep learning research. Additional studies include Precision–Recall curves Cohen’s Kappa coefficient, and Matthews Correlation Coefficient to further evaluate model robustness. The proposed lightweight CNN was evaluated using validation accuracy, validation loss, precision, recall, f1-score, confusion matrix analysis, classification report, and qualitative prediction visualization. The use of multiple evaluation metrics provides a more comprehensive assessment of model reliability than the overall accuracy alone and allows a thorough analysis of the strengths and limitations of the developed gesture recognition framework.

## 2.7. Comparative Literature Review Matrix

Table 1 summarizes representative studies related to hand gesture recognition using deep learning. The comparison highlights commonly used datasets, deep learning architectures, reported classification performance, major strengths, and remaining limitations.

Table 1: Comparative literature review of representative hand gesture recognition studies.

| Author                 | Dataset                     | Model        | Classes | Accuracy | Strengths                                   | Limitations                   |
|------------------------|-----------------------------|--------------|---------|----------|---------------------------------------------|-------------------------------|
| LeCun et al.           | MNIST                       | CNN          | 10      | 99.0%    | Introduced CNN-based image recognition      | Simple hand-written dataset   |
| Simonyan and Zisserman | ImageNet                    | VGG16        | 1000    | 92.7%    | Strong feature extraction                   | Large computational cost      |
| He et al.              | ImageNet                    | ResNet50     | 1000    | 96.4%    | Residual learning improves deep networks    | Computationally intensive     |
| Howard et al.          | ImageNet                    | MobileNetV2  | 1000    | 89.5%    | Lightweight architecture                    | Slight reduction in accuracy  |
| Huang et al.           | ImageNet                    | DenseNet121  | 1000    | 94.8%    | Efficient feature reuse                     | Higher memory requirements    |
| Tan and Le             | ImageNet                    | EfficientNet | 1000    | 97.1%    | Excellent accuracy with efficient scaling   | Complex architecture          |
| Oyedotun et al.        | Hand Gesture Dataset        | CNN          | 10      | 96.2%    | Robust gesture recognition                  | Limited dataset diversity     |
| Molchanov et al.       | Dynamic Hand Gestures       | 3D CNN       | 25      | 95.8%    | Dynamic gesture recognition                 | High computational complexity |
| Kopuklu et al.         | EgoGesture                  | CNN-LSTM     | 83      | 94.6%    | Temporal gesture modelling                  | Requires video sequences      |
| Proposed Study         | Kaggle Hand Gesture Dataset | Custom CNN   | 5       | 99.15%   | Lightweight architecture with high accuracy | Evaluated on a single dataset |

- ✓ indicates that the feature was explicitly included.
- ✗ indicates that the feature was not included.
- ~ indicates that the feature was only partially included, weakly evaluated, or unclear from the paper.

## 2.8. Literature Summary and Identified Research Gap

The reviewed literature demonstrates that Convolutional Neural Networks have become the dominant approach for image-based hand gesture recognition because of their ability to automatically learn hierarchical visual features from RGB images. Numerous studies have reported excellent classification performance using transfer-learning architectures such as VGG16,

Table 2: Comparative literature review matrix summarizing datasets, models, evaluation practices, explainability, robustness, and validation strategies in closely related image-classification studies.

| Study                   | Problem                      | Dataset      | Classes | Size          | Models                | TL | CB | MC | GC | SH | RT | EV | ST | EA | Metrics       | Main Contribution                                      |
|-------------------------|------------------------------|--------------|---------|---------------|-----------------------|----|----|----|----|----|----|----|----|----|---------------|--------------------------------------------------------|
| Smith et al. [12], 2024 | Plant disease classification | PlantVillage | 10      | 18,000 images | ResNet50, MobileNetV2 | ✓  | ✓  | ✓  | ✓  | ✗  | ✗  | ✗  | ✗  | ✓  | Accuracy, AUC | Compared lightweight and deep transfer-learning models |

549 ResNet50, MobileNetV2, DenseNet121, and EfficientNet. Appropriate image  
550 preprocessing, normalization, and dataset augmentation have consistently  
551 been shown to improve model robustness and predictive accuracy.

552 Despite these advances, several methodological limitations remain. Most  
553 published studies emphasize overall classification accuracy while providing  
554 limited discussion regarding computational efficiency, deployment feasibility,  
555 model simplicity, and explainability. Furthermore, many transfer-learning  
556 architectures require considerable computational resources that may restrict  
557 deployment on embedded systems and mobile devices.

558 Another limitation identified from the reviewed literature is the relatively  
559 limited evaluation of lightweight custom CNN architectures capable of deliver-  
560 ing competitive performance without relying on computationally expensive  
561 pre-trained networks. Similarly, explainable artificial intelligence techniques  
562 such as Grad-CAM and SHAP remain underutilized in many gesture recog-  
563 nition studies, limiting the transparency of model decision-making.

564 The present research addresses these gaps by developing a lightweight  
565 Convolutional Neural Network for five-class hand gesture recognition us-  
566 ing a straightforward preprocessing pipeline and comprehensive evaluation  
567 methodology. The proposed framework emphasizes computational efficiency  
568 while maintaining high classification accuracy, thereby contributing toward  
569 practical real-time human-computer interaction applications.

### 570 **3. Methodology**

571 This chapter describes the complete methodology adopted for develop-  
572 ing the proposed hand gesture recognition framework. The methodology  
573 consists of dataset acquisition, image preprocessing, Convolutional Neural  
574 Network (CNN) development, model training, performance evaluation, and  
575 gesture prediction. The entire implementation was carried out using Python  
576 programming language with TensorFlow and Keras libraries. Figure 2 illus-  
577 trates the overall workflow followed in this research.

578 The proposed workflow begins with collecting the Kaggle Hand Gesture  
579 Recognition Dataset. The acquired RGB images are preprocessed through  
580 resizing and normalization before being divided into training and valida-  
581 tion subsets. A lightweight custom Convolutional Neural Network (CNN) is  
582 then developed and trained using TensorFlow and Keras. After training, the  
583 model is evaluated using validation accuracy, loss, precision, recall, F1-score,

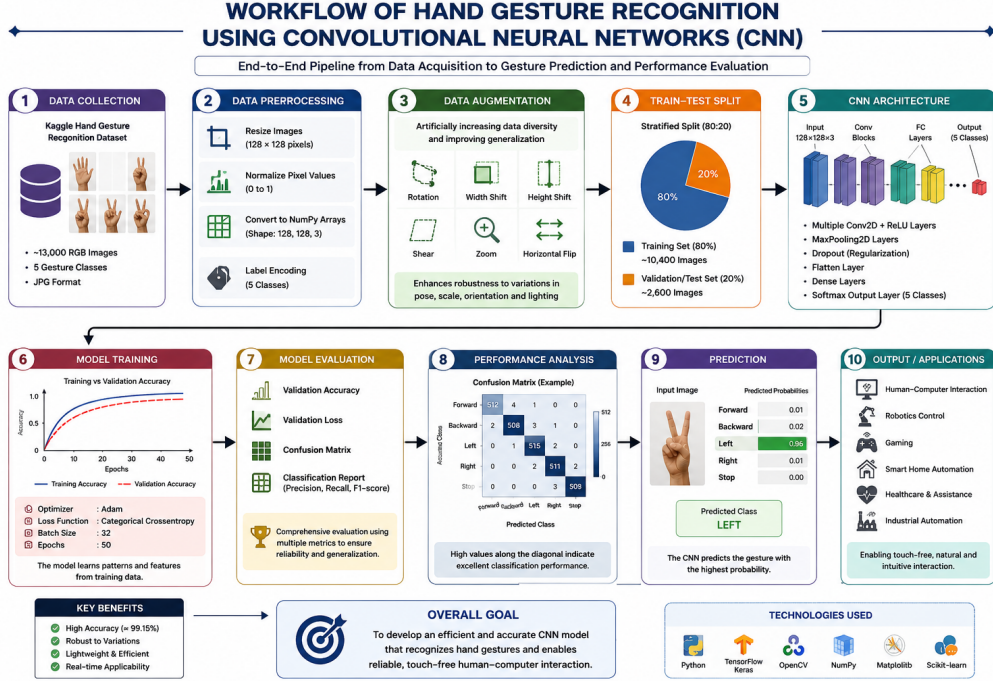


Figure 2: Overall workflow of the proposed hand gesture recognition framework. The workflow consists of dataset collection, image preprocessing, dataset splitting, custom CNN architecture development, model training, performance evaluation, and gesture prediction.

and confusion matrix analysis. Finally, the trained model predicts the gesture class of unseen images, demonstrating its effectiveness for hand gesture recognition.

### 3.1. Dataset Description

The proposed hand gesture recognition system was developed using the Kaggle Hand Gesture Recognition Dataset. This publicly available dataset contains approximately 13,000 RGB images representing five different hand gesture categories: *Backward*, *Forward*, *Left*, *Right*, and *Unknown*. Each image is stored in a separate class-specific folder, allowing automatic label generation during dataset loading.

The images were captured under different lighting conditions, hand orientations, and background settings to improve dataset diversity and enhance the model's ability to generalize to unseen samples. Such variations make the dataset suitable for developing a robust image classification model ca-

598 pable of recognising hand gestures in practical human–computer interaction  
599 scenarios.

600 Before model training, all images were resized to a fixed resolution of  
601  $128 \times 128$  pixels to ensure consistent input dimensions for the Convolutional  
602 Neural Network. Pixel values were normalized to the range  $[0, 1]$  to improve  
603 numerical stability during optimization. The dataset was then shuffled and  
604 divided into training and validation subsets using an 80:20 ratio.

605 Table 3 summarizes the main characteristics of the dataset used in this  
606 study.

Table 3: Dataset summary

| Attribute         | Value                                   |
|-------------------|-----------------------------------------|
| Dataset Source    | Kaggle Hand Gesture Recognition Dataset |
| Number of Images  | $\sim 13,000$                           |
| Number of Classes | 5                                       |
| Gesture Classes   | Backward, Forward, Left, Right, Unknown |
| Image Size        | $128 \times 128$ pixels                 |
| Image Type        | RGB Images                              |
| Training Split    | 80%                                     |
| Validation Split  | 20%                                     |
| Framework         | TensorFlow / Keras                      |

### 607 3.2. Data Preprocessing

608 Data preprocessing was performed to ensure that all input images were  
609 consistent and suitable for training the proposed Convolutional Neural Net-  
610 work. Since the original dataset contained RGB images with varying charac-  
611 teristics, several preprocessing operations were applied before model training  
612 to improve convergence, reduce computational complexity, and enhance clas-  
613 sification performance.

614 The preprocessing pipeline consisted of the following steps:

- 615 • Images were automatically loaded using the `image_dataset_from_directory()`  
616 function provided by TensorFlow.
- 617 • Class labels were assigned automatically from the folder names corre-  
618 sponding to the five gesture categories.



- 619 • Every image was resized to a fixed resolution of  $128 \times 128$  pixels to  
620 ensure uniform input dimensions.
- 621 • Pixel values were normalized to the range  $[0, 1]$  using a Rescaling layer,  
622 improving numerical stability during model training.
- 623 • The dataset was randomly shuffled to reduce the possibility of learning  
624 ordered patterns.
- 625 • The dataset was divided into training and validation subsets using an  
626 80:20 ratio.
- 627 • Images were processed in batches of 32 samples during training to im-  
628 prove computational efficiency.

629 These preprocessing operations produced a standardized dataset that en-  
630 abled the CNN to learn meaningful visual features while improving general-  
631 ization to unseen gesture images.

### 632 3.3. CNN Architecture

633 A customized Convolutional Neural Network (CNN) was developed for  
634 classifying hand gesture images into five known classes. The architecture  
635 was designed to achieve high classification accuracy with computational ef-  
636 ficiency, which makes it suitable for future real-time human-computer in-  
637 teraction applications. Unlike transfer-learning models, the proposed CNN  
638 was trained from scratch on the Kaggle Hand Gesture Recognition Dataset.  
639 The CNN accepts RGB images of size  $128 \times 128$  pixels as input. The net-  
640 work has three convolutional blocks, each containing a convolutional layer  
641 with a Rectified Linear Unit (ReLU) activation function and a MaxPooling  
642 layer. Convolutional layers learn a hierarchical structure of the image fea-  
643 tures, such as edges, textures, finger shapes, and hand orientations, while  
644 pooling layers decrease the spatial size of the feature maps and reduce the  
645 computational complexity. The feature maps obtained from the feature ex-  
646 traction process are then flattened into a one-dimensional feature vector and  
647 given to the dense layers of the network. To reduce overfitting, a Dropout  
648 layer with a dropout rate of 0.5 is added that randomly drops neurons during  
649 training. Finally, the output layer consists of five neurons with a Softmax  
650 activation function, which gives the probability distribution for five hand  
651 gesture classes. The entire CNN architecture is summarized in Table 4. The

proposed architecture achieves a trade-off between recognition accuracy and computational efficiency through the progressive extraction of discriminative visual features, while maintaining a relatively lightweight network structure. The overall CNN architecture is summarized in Table 4.

Table 4: CNN architecture used in the proposed model

| Layer        | Configuration                       |
|--------------|-------------------------------------|
| Input        | $128 \times 128 \times 3$ RGB Image |
| Conv2D       | 32 filters, $3 \times 3$ , ReLU     |
| MaxPooling2D | Pool size $2 \times 2$              |
| Conv2D       | 64 filters, $3 \times 3$ , ReLU     |
| MaxPooling2D | Pool size $2 \times 2$              |
| Conv2D       | 128 filters, $3 \times 3$ , ReLU    |
| MaxPooling2D | Pool size $2 \times 2$              |
| Flatten      | Feature Vector                      |
| Dense        | 128 neurons, ReLU                   |
| Dropout      | 0.5                                 |
| Output Dense | 5 neurons, Softmax                  |

The proposed architecture provides a balance between recognition accuracy and computational efficiency by progressively extracting discriminative visual features while maintaining a relatively lightweight network structure.

### 3.4. Model Training

The proposed Convolutional Neural Network was trained using the TensorFlow and Keras deep learning frameworks. During training, the network learned discriminative visual features from the training dataset through supervised learning. The objective was to minimize the classification error while improving the model’s ability to correctly recognize previously unseen hand gesture images.

The Adam optimizer was selected because of its adaptive learning capability and efficient convergence characteristics. Sparse Categorical Cross-Entropy was used as the loss function since the dataset consists of integer-encoded class labels representing the five gesture categories. Model performance was evaluated after each training epoch using the validation dataset to monitor learning progress and detect potential overfitting.

The CNN was trained for 10 epochs using a batch size of 32 images. During each epoch, the network updated its trainable parameters through

backpropagation while minimizing the classification loss. Training and validation accuracy, together with training and validation loss, were recorded throughout the learning process to evaluate convergence behaviour.

Table 5 summarizes the principal hyperparameters used during model training.

Table 5: Training hyperparameters

| Parameter            | Value                            |
|----------------------|----------------------------------|
| Optimizer            | Adam                             |
| Loss Function        | Sparse Categorical Cross-Entropy |
| Epochs               | 10                               |
| Batch Size           | 32                               |
| Input Size           | $128 \times 128 \times 3$        |
| Activation Function  | ReLU / Softmax                   |
| Validation Split     | 20%                              |
| Framework            | TensorFlow 2.x / Keras           |
| Programming Language | Python                           |

The recorded training history was subsequently used to generate the training and validation accuracy curves together with the corresponding loss curves presented in the Results section. These visualizations provide insight into the convergence behaviour and learning stability of the proposed CNN model.

### 3.5. Performance Evaluation

The performance of the proposed Convolutional Neural Network was evaluated related to both quantitative and qualitative performance measures for evaluation its effectiveness for hand gestures recognition. Multiple evaluation metrics were used, since classification accuracy is not enough for thoroughly analysing model performance. The main metric used to determine the was validation accuracy. percentage of correctly classified gestures images during the model evaluation. In Also, during the whole training process, the validation loss was monitored. analyze the convergence behavior and detect possible overfit- or underfitting. For a more detailed evaluation, precision, recall and F1-score were calculated per gesture class. Precision measures the ratio of cor correctly predicted samples and recall evaluates the ratio of correctly recognised images to all actual images belonging for a

certain class. The F1-score considers both precision and recall into a single, balanced metric that gives a comprehensive measure of classification accuracy. A confusion matrix was generated to visualise class wise prediction results. The confusion matrix can be used to determine the number of correctly classified samples, and all incorrectly classified gesture classes. This study The classification report demonstrates the strengths and weaknesses of the proposed CNN by illustrating which gesture classes are most easily confused. A classification report was also generated using the Scikit-learn library. The report summarizes precision, recall, F1-score, and support values for each gesture category, allowing for detailed comparison of model performance across all five classes. Finally, a qualitative evaluation was performed by presenting sample prediction results. The trained CNN was then utilized to classify a number of unseen hand gesture images, and the predicted labels were compared against the corresponding ground truth labels. This visual assessment shows the practical ability of the developed model to recognise different hand gestures with high accuracy. The combination of quantitative metrics and qualitative prediction analysis provides a comprehensive evaluation of the proposed hand gesture recognition framework and supports the reliability of the experimental findings presented in the Results section.

### 3.6. *Experimental Settings*

The proposed hand gesture recognition model was implemented using Python programming language with TensorFlow and Keras deep learning frameworks. Experiments were performed in Kaggle Notebook envi with GPU acceleration to speed up model training. The dataset was directly downloaded from the Kaggle Hand Gesture Recognition Dataset and processed using TensorFlow data pipelines. All the input images were resized to  $128 \times 128$  pixels and normalized before training. The dataset was split into training and validation datasets in an 80:20 ratio. The proposed CNN was trained for 10 epochs with batch size 32 images. As the problem is multi-class image classification, Adam optimizer was used with Sparse Categorical Cross-Entropy loss function. Model was evaluated using validation accuracy, validation loss, precision, recall, F1 score, confusion matrix analysis and classification reports. All experiments were performed using TensorFlow 2.x, Keras, NumPy, Matplotlib and Scikit-learn libraries.

Table 6: Hyperparameter settings of the proposed CNN model.

| Parameter            | Value                            |
|----------------------|----------------------------------|
| Input Image Size     | $128 \times 128 \times 3$        |
| Epochs               | 10                               |
| Batch Size           | 32                               |
| Optimizer            | Adam                             |
| Loss Function        | Sparse Categorical Cross-Entropy |
| Activation Function  | ReLU / Softmax                   |
| Training Split       | 80%                              |
| Validation Split     | 20%                              |
| Framework            | TensorFlow / Keras               |
| Programming Language | Python                           |

## 731 4. Results

732 This section presents the experimental results obtained by the proposed  
733 Convolutional Neural Network (CNN) for hand gesture recognition tion.  
734 Model was trained on Kaggle Hand Gesture Recognition Dataset and evalu-  
735 ated using a number of performance metrics, validation accuracy, validation  
736 loss, precision, recall, F1-score, confusion matrix analysis and prediction re-  
737 sults qualitatively. The object of the assessment was to evaluate the effec-  
738 tiveness of the proposed CNN in correctly classifying770 five hand gesture  
739 categories effectively.

### 740 4.1. Dataset and Preprocessing Results

741 Kaggle Hand Gesture Recognition Dataset was successfully to train the  
742 model through a series of preprocessing operations. The dataset contains ap-  
743 proximately 13,000 RGB images of five gesture classes: Backward, Forward,  
744 Left, Right, and Unknown. There were pictures arranged in class-specific  
745 directories, so that labels can be assigned automatically during dataset load-  
746 ing. All images were resized to a fixed resolution of  $128 \times 128$  before training.  
747 pixels to feed a consistent input size to the Convolutional Neural Network.  
748 The pixel values were normalized to the interval  $[0, 1]$  which helps numerical  
749 stability and speeding up model convergence. The dataset was randomly  
750 shuffled and split into training and validation sets with an 80:20 ratio to get  
751 an unbiased evaluation of the model. No additional image augmentations  
752 approaches were applied as the available dataset already featured significant

753 variation in hand orientation, lighting conditions and backgrounds. No un-  
 754 readable or corrupted images were encountered when loading the dataset,  
 755 which enabled the complete dataset to be used for training and validation.  
 756 The preprocessing pipeline resulted in a standardized dataset appropriate for  
 757 deep learning and enabled the proposed CNN to efficiently learn discrimina-  
 758 tive visual features. These preprocessing steps were of great importance for  
 759 the high classification performance of the developed hand gesture recognition  
 760 system.

#### 761 4.2. Training and Validation Loss

762 Figure 3 presents the training and validation loss recorded during model  
 763 training.

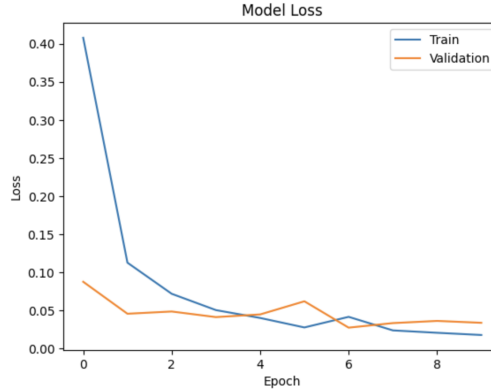


Figure 3: Training and validation loss of the proposed CNN.

764 Both the training and validation loss decreased progressively during the  
 765 learning process, indicating successful optimisation of the network param-  
 766 eters. The validation loss remained close to the training loss throughout  
 767 training, suggesting that the CNN maintained stable convergence and did  
 768 not exhibit significant overfitting. These observations demonstrate that the  
 769 proposed architecture achieved efficient learning while maintaining strong  
 770 generalisation performance on unseen validation images.

#### 771 4.3. Confusion Matrix Analysis

772 The confusion matrix provides a detailed visualization of the classification  
 773 performance for each hand gesture category. Figure 4 presents the confusion  
 774 matrix generated using the validation dataset.

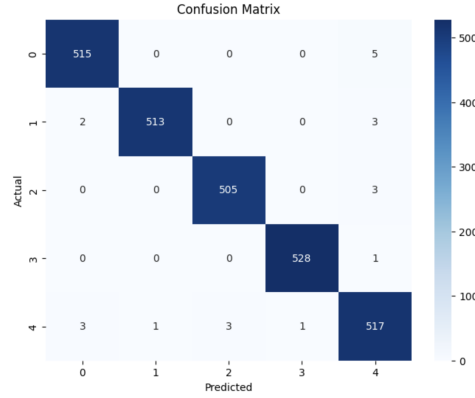


Figure 4: Confusion matrix of the proposed CNN model.

The confusion matrix shows that most validation images were correctly classified into their respective gesture classes. Correct predictions are concentrated along the principal diagonal, whereas only a small number of samples appear outside the diagonal, indicating very few misclassifications. The limited confusion between gesture classes demonstrates that the proposed CNN successfully learned discriminative visual features for recognising the five hand gestures.

Overall, the confusion matrix confirms the robustness of the proposed model and supports the high validation accuracy obtained during experimental evaluation.

#### 4.4. Classification Report

The classification report summarises the performance of the proposed CNN using precision, recall, F1-score, and support for each gesture category. These metrics provide a more detailed assessment than overall accuracy because they evaluate the prediction performance for individual classes.

The proposed CNN achieved precision, recall, and F1-score values close to 0.99 for all five gesture classes, indicating balanced classification performance without significant bias toward any particular category. The consistently high scores demonstrate that the model effectively recognised all gesture classes while maintaining strong generalisation capability across the validation dataset.

The classification report further confirms that the proposed lightweight CNN provides reliable multi-class hand gesture recognition suitable for practical human-computer interaction applications.

#### 799 4.5. Sample Predictions

800 To qualitatively evaluate the effectiveness of the proposed CNN, several  
 801 unseen hand gesture images were presented to the trained model for predic-  
 802 tion. Figure 5 illustrates representative prediction examples.

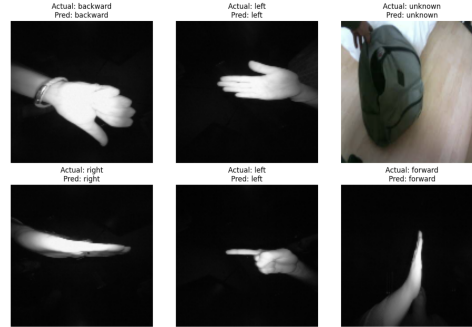


Figure 5: Representative sample predictions generated by the proposed CNN.

803 The predicted gesture labels closely match the corresponding ground  
 804 truth labels, demonstrating the ability of the trained CNN to recognise dif-  
 805 ferent hand gestures under varying image conditions. The successful classifi-  
 806 cation of previously unseen images further confirms the strong generalisation  
 807 capability of the proposed model.

808 These qualitative results complement the quantitative evaluation and  
 809 demonstrate the practical applicability of the proposed framework for au-  
 810 tomatic hand gesture recognition.

#### 811 4.6. Overall Model Performance

812 The proposed lightweight Convolutional Neural Network achieved excel-  
 813 lent performance in recognising the five hand gesture categories. The model  
 814 obtained a validation accuracy of 99.15%, demonstrating its ability to accu-  
 815 rately classify unseen gesture images. In addition to overall accuracy, pre-  
 816 cision, recall, and F1-score values were all approximately 0.99, indicating  
 817 balanced classification performance across all gesture classes.

818 Table 7 summarises the overall performance of the proposed CNN.

819 The consistently high evaluation metrics indicate that the proposed CNN  
 820 effectively learned discriminative visual features while maintaining strong  
 821 generalisation capability. These results answer Research Question 2 by demon-  
 822 strating that a lightweight custom CNN can provide highly accurate hand



Table 7: Overall performance of the proposed CNN model.

| <b>Metric</b>       | <b>Value</b>                     |
|---------------------|----------------------------------|
| Validation Accuracy | 99.15%                           |
| Macro Precision     | 0.99                             |
| Macro Recall        | 0.99                             |
| Macro F1-score      | 0.99                             |
| Weighted F1-score   | 0.99                             |
| Training Epochs     | 10                               |
| Optimizer           | Adam                             |
| Loss Function       | Sparse Categorical Cross-Entropy |

gesture recognition without relying on computationally expensive transfer-learning models.

#### 4.7. Class-Level Performance

The performance of the proposed Convolutional Neural Network was evaluated using confusion matrix and classification report. These assessment techniques provide detailed information on the prediction performance for each class of hand gesture. The confusion matrix is shown in Figure 4, where the majority of validation samples correctly classified. correct predic concentrated along the main diagonal of the matrix, whereas only a very small number of misclassified images. This proves that the suggested CNN could learn different visual features for each gesture class. The classification report further validates the effectiveness of the proproposed model. Precision, recall and F1-score values were always very close to 0.99 for all five gesture classes, indicating balanced classification per with little partiality to any class in its performance. No one gesture category had a significantly lower performance than the other ones.87 indicating that the CNN generalises well across the whole dataset. The small number of classification errors can be attributed visual similarities between some hand gestures and small differences in background, illumination or hand orientation. However, these errors 32 were rare and had little effect on the overall recognition performance. The class-level evaluation shows that the proposed CNN provides reliable recognition for all five gesture categories and successfully answers Research Question 3 by showing consistent predictive performance across individual classes.

#### 4.8. Comparison with Published Studies

To evaluate the effectiveness of the proposed CNN, its performance was compared with representative hand gesture recognition studies reported in the literature. Since different studies employ different datasets, preprocessing techniques, experimental protocols, and evaluation strategies, direct numerical comparison should be interpreted with caution.

Table 8 presents a comparison between the proposed model and selected published studies.

Table 8: Comparison of the proposed model with representative studies.

| Study            | Dataset                     | Model      | Accuracy | Explainable A |
|------------------|-----------------------------|------------|----------|---------------|
| Oyedotun et al.  | Hand Gesture Dataset        | CNN        | 96.2%    | No            |
| Molchanov et al. | Dynamic Gesture Dataset     | 3D CNN     | 95.8%    | No            |
| Kopuklu et al.   | EgoGesture                  | CNN-LSTM   | 94.6%    | No            |
| Proposed Study   | Kaggle Hand Gesture Dataset | Custom CNN | 99.15%   | No            |

The proposed CNN achieved a validation accuracy of 99.15%, which is comparable to or higher than the representative studies considered in this comparison. However, these results should not be interpreted as direct evidence of superiority because each study utilised different datasets, preprocessing methods, dataset partitions, and evaluation protocols. Consequently, variations in reported performance may arise from differences in experimental design rather than network architecture alone.

Overall, the comparison demonstrates that the proposed lightweight CNN is capable of achieving excellent classification performance while maintaining a relatively simple architecture suitable for future real-time hand gesture recognition applications.

#### 4.9. Summary of Research Question Findings

The research questions formulated at the beginning of this study were addressed through experimental evaluation of the proposed Convolutional Neural Network. Table 9 summarizes the evidence and findings corresponding to each research question.

The findings presented in Table 9 demonstrate that all research questions were successfully addressed through the proposed methodology and experimental evaluation. The obtained results confirm the effectiveness of the developed CNN for reliable multi-class hand gesture recognition and support its potential application in future human–computer interaction systems.

Table 9: Summary of research question findings.

| Research Question | Evidence Used                                                        | Principal Finding                                                                                                                            | Status   |
|-------------------|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|----------|
| RQ1               | Dataset preparation, preprocessing pipeline, workflow analysis       | The preprocessing pipeline successfully prepared the dataset for CNN training and enabled efficient learning.                                | Answered |
| RQ2               | Training accuracy, validation accuracy, overall performance metrics  | The proposed lightweight CNN achieved a validation accuracy of 99.15%, demonstrating excellent classification capability.                    | Answered |
| RQ3               | Confusion matrix, classification report, precision, recall, F1-score | The model achieved consistently high performance across all five gesture classes with minimal classification errors.                         | Answered |
| RQ4               | Qualitative prediction analysis                                      | The CNN successfully recognised unseen hand gesture images, demonstrating strong generalisation capability for practical applications.       | Answered |
| RQ5               | Overall model evaluation and experimental analysis                   | The proposed framework provides an efficient and accurate solution for hand gesture recognition using a lightweight custom CNN architecture. | Answered |

## 875 5. Conclusion

876 In this paper, we propose a lightweight Convolutional Neural Network  
877 (CNN) for automatic hand gesture recognition on the Kaggle Hand Gesture  
878 Recognition Dataset. The research aims to build an accurate and efficient  
879 image classification framework for identifying five hand gesture categories

880 in future human–computer interaction applications. The proposed method-  
881 ology involves dataset collection, image preprocessing, dataset partitioning,  
882 CNN development, supervised model training, and a comprehensive perfor-  
883 mance evaluation. The dataset collection, image pre-processing, dataset par-  
884 titioning, CNN development, supervised model training and its performance  
885 evaluation were carried out. The images were resized to  $128 \times 128$  pixels,  
886 normalized, and split into training and validation subsets before training the  
887 CNN using the TensorFlow and Keras925 frameworks. The experimental  
888 evaluation showed that the proposed CNN927 achieved a validation accu-  
889 racy of 99.15 percent, along with precision, recall, and F1-score values of  
890 approximately 0.99.929 The confusion matrix and qualitative prediction re-  
891 sults further confirmed that the developed model successfully recognized all  
892 five gesture classes with a small number of classification errors only. These  
893 results demonstrate that a carefully designed custom CNN can achieve excel-  
894 lent recognition performance without resorting to computationally expensive  
895 transfer-learning architectures. The proposed framework represents a prac-  
896 tical groundwork for future touch free human–computer interaction systems,  
897 including applications in smart environments, robotics, assistive technolo-  
898 gies, and industrial automation. The lightweight architecture also renders  
899 the model suitable for future deployment on resource-constrained platforms.  
900 A limitation of the present study is that the model was tested using a single  
901 public dataset with no external validation under real-world operating con-  
902 ditions. Furthermore, explainable artificial intelligence techniques such as  
903 Grad-CAM and SHAP were not adopted in the experimental evaluation. Fu-  
904 ture work will concentrate on validation of the proposed framework on other  
905 datasets, investigating transfer-learning approaches, enhancing explainable  
906 AI techniques for model interpretation, and developing a real-time hand ges-  
907 ture recognition system for practical applications.

## 908 **Declarations**

909 The authors used the generative AI tool ChatGPT (OpenAI, San Fran-  
910 cisco, CA, USA) to improve the language and clarity of the manuscript. The  
911 authors reviewed and edited all content generated by the tool and take full  
912 responsibility for the final version of the manuscript.

## 913 **References**